Design and Evaluation of a Digital Quality Assurance System for Learning Outcomes in Kazakhstan's Higher Education Sector

Conceptual and Methodological Justification and Financial Justification

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1. Project Title

Design and Evaluation of a Digital Quality Assurance System for Learning Outcomes in Kazakhstan's Higher Education Sector

2. Abstract

Abstract (569 words)

Problem

Higher education institutions are adopting learning management systems, electronic assessment tools, and digital repositories, yet evidence about course and program quality often remains fragmented across platforms, spreadsheets, and narrative reports. This creates an unresolved contradiction: institutions collect more digital data than before, but reviewers still lack a consistent, timely, and auditable way to connect assessment evidence with learning outcomes and improvement decisions. As a result, quality assurance can become a periodic reporting exercise instead of a continuous process that supports teaching.

Goal

By December 2028, the project will design, develop, and pilot a digital quality assurance workflow in three Kazakhstan higher education institutions, applying it to at least 600 course assessment packages and evaluating it against baseline review practice. The evaluation will measure reviewer agreement, completeness of evidence mapping, and review turnaround time, with target thresholds of Cohen's kappa of at least 0.70, at least 90 percent completeness, and a 20 percent reduction in median turnaround time.

Methods

The study will use a mixed-methods, non-randomized before-and-after design. A standards review and process mapping will define the evidence model; co-design with quality assurance staff, instructors, and program leaders will inform a secure prototype and user guide. At each institution, a stratified sample of 200 course packages will include 100 from the prior manual review cycle and 100 prospectively reviewed using the workflow, matched by program where feasible. Two trained reviewers will independently apply a categorical evidence-sufficiency rubric to a common stratified subset of at least 120 packages (40 per institution); Cohen's kappa will measure agreement. Completeness will be the share of required outcome-assessment links with traceable evidence. Review time will span receipt of a complete package to sign-off. Comparisons will report uncertainty and institution and discipline differences; because they are non-randomized, results will not support unqualified causal claims. Usability interviews, security, data minimization, de-identification, and institutional approval will inform responsible deployment; software organizes evidence and does not replace academic judgment.

Expected Results

The project is expected to produce a validated conceptual model, a working prototype, a documented pilot dataset and evaluation report, and recommendations for responsible institutional adoption. It will test whether a common evidence structure improves the traceability and consistency of course review, while identifying which outcomes depend on local policy, reviewer training, or system integration. The targets in the goal are evaluation criteria, not assumed findings. If the pilot does not meet a threshold, the report will document the result, uncertainty, and reasons rather than claim success.

Practical Significance

The prototype and guide will help quality assurance offices move from manually assembled evidence toward a repeatable review process. Program leaders will see which outcomes are supported by assessments, where evidence is missing, and which actions were agreed. The project will provide a data dictionary, workflow templates, training materials, a technical specification, and an adaptable report. Expansion will depend on local governance, accessibility, and compatibility with existing learning platforms.

Level of Impact

The immediate impact is institutional: participating programs can test a more consistent way to prepare and review evidence. The project aims for a methodological contribution by specifying measurable indicators and a pilot protocol that other institutions can reproduce. A wider national impact is conditional on independent validation, sustainable maintenance, policy alignment, and further funding. Its contribution will be evidence about feasibility, measurement quality, usability, and the conditions under which digital quality assurance can support decisions.

3. Problem

Although higher education institutions have expanded digital learning and assessment, quality assurance evidence remains distributed across incompatible systems and manual reports. The unresolved contradiction is that greater data availability has not produced a more continuous, comparable, or auditable connection between learning outcomes, assessment evidence, and improvement decisions.

4. SMART Goal

By 31 December 2028, design, implement, and evaluate a digital quality assurance workflow at three Kazakhstan higher education institutions using at least 600 course assessment packages, and determine whether it achieves reviewer agreement of at least 0.70 Cohen's kappa, evidence-mapping completeness of at least 90 percent, and a 20 percent reduction in median review time against baseline practice.

5. Objectives

1. Analyze international and Kazakhstan higher education quality assurance approaches and map the evidence, roles, and data flows used in course and program review.
2. Develop a measurable evidence model linking learning outcomes, assessment tasks, rubrics, review judgments, and improvement actions; define indicators and a pre-specified evaluation protocol.
3. Co-design and implement a secure prototype and user guidance with quality assurance staff, instructors, and program leaders, then refine it through usability and technical checks.
4. Pilot the workflow in three institutions on at least 600 course assessment packages, collecting comparable baseline and digital-process measures under approved data governance.
5. Analyze reviewer agreement, evidence completeness, review time, usability, and implementation differences; publish findings, limitations, and adoption recommendations.

6. Hypothesis

Compared with matched prior-cycle manual reviews, the digital workflow will achieve at least 90 percent completeness (required outcome-assessment links with traceable evidence divided by all required links) and at least 20 percent lower median review time (from receipt of a complete package to sign-off); two trained reviewers applying the categorical evidence-sufficiency rubric will achieve Cohen's kappa of at least 0.70 on a common sample of 120 or more packages. Because allocation is non-randomized, the analysis will report observed differences with institution and discipline uncertainty rather than claim unqualified causal effects.

7. Expected Results

- A documented quality assurance evidence model, indicator dictionary, and pre-specified evaluation protocol, corresponding to Objectives 1 and 2.
- A tested prototype, technical specification, user guide, and training package, corresponding to Objective 3.
- A de-identified pilot dataset and evaluation report covering 600 or more course assessment packages across three institutions, corresponding to Objective 4.
- At least one manuscript submitted to a peer-reviewed education or educational technology journal and a concise institutional policy brief presenting results, uncertainty, and conditions for adoption, corresponding to Objective 5.

8. Financial Justification

Financial analysis (379 words)

The estimate is retained in the units and amounts supplied: thousand tenge. Labor costs total 51,215,424, or about 60.6 percent of the 84,574,008 thousand tenge envelope. The 2026 allocation is 7,249,632 for standards analysis, consultation, and evidence-model design before prototype development and piloting. Labor rises to 21,936,096 in 2027 and 22,029,696 in 2028 as software developers, data and evaluation specialists, accessibility and security staff, trainers, and institutional coordinators build, deploy, support, and analyze the pilot. These annual commitments should be tied to named roles, effort, and deliverables.

International business trips total 6,144,584: 2,006,212 in 2026, 2,132,160 in 2027, and 2,006,212 in 2028. The near-even distribution supports early standards consultation, mid-project exchange on prototype and pilot design, and final dissemination. Each trip needs a defined purpose, costed itinerary, participants, and report; use remote meetings where adequate.

Other services and works total 16,700,000, phased at 7,500,000, 4,100,000, and 5,100,000. The larger 2026 amount can cover requirements and interface design, security and accessibility assessment, and prototype setup. The 2027 allocation supports integration, user testing, and independent evaluation during piloting. The 2028 amount covers maintenance, fixes, documentation, and handover after evaluation. Procurement should specify outputs and acceptance criteria to avoid overlap with salaried labor.

Materials are 14,000 overall (1,000, 4,000, and 9,000 by year) for modest consumables, printing of training or consent materials where necessary, and pilot logistics; digital distribution should keep this line low. Equipment and software total 10,500,000, declining from 5,500,000 in 2026 to 3,000,000 in 2027 and 2,000,000 in 2028. Initial investment establishes the shared training and secure testing environment; later purchases cover capacity additions, subscriptions, and licenses. Line-item year sums are 22,256,844, 31,172,256, and 31,144,908; the supplied 2028 total and three stated annual totals exceed the grand total by 5,348 thousand tenge. Reconcile this discrepancy before approval. At the stated unit, the total is 84.574 trillion tenge. This three-institution demonstrator cannot credibly absorb that envelope, and the brief supplies no staff quantities, quotations, or rollout schedule to support it. Treat the figures as an unverified program-level estimate: before approval, the sponsor must confirm whether the unit is truly thousand tenge or provide the national rollout scope and itemized cost basis. The category and phase rationales below do not validate the total.

Consolidated estimated cost statement (thousand tenge)

Expenditure item	Total	2026	2027	2028
Labor Costs (incl. taxes and mandatory payments)	51,215,424	7,249,632	21,936,096	22,029,696
Business Trips (outside the Republic of Kazakhstan)	6,144,584	2,006,212	2,132,160	2,006,212
Other Services and Works	16,700,000	7,500,000	4,100,000	5,100,000
Acquisition of Materials	14,000	1,000	4,000	9,000
Acquisition of Equipment and/or Software	10,500,000	5,500,000	3,000,000	2,000,000
Total	**84,574,008**	**22,256,844**	**31,172,256**	**31,150,256**

Budget reconciliation: The supplied 2028 total (31,150,256) is 5,348 thousand tenge higher than its line-item sum (31,144,908). The supplied annual totals sum to 84,579,356, while the grand total and sum of line items are 84,574,008.